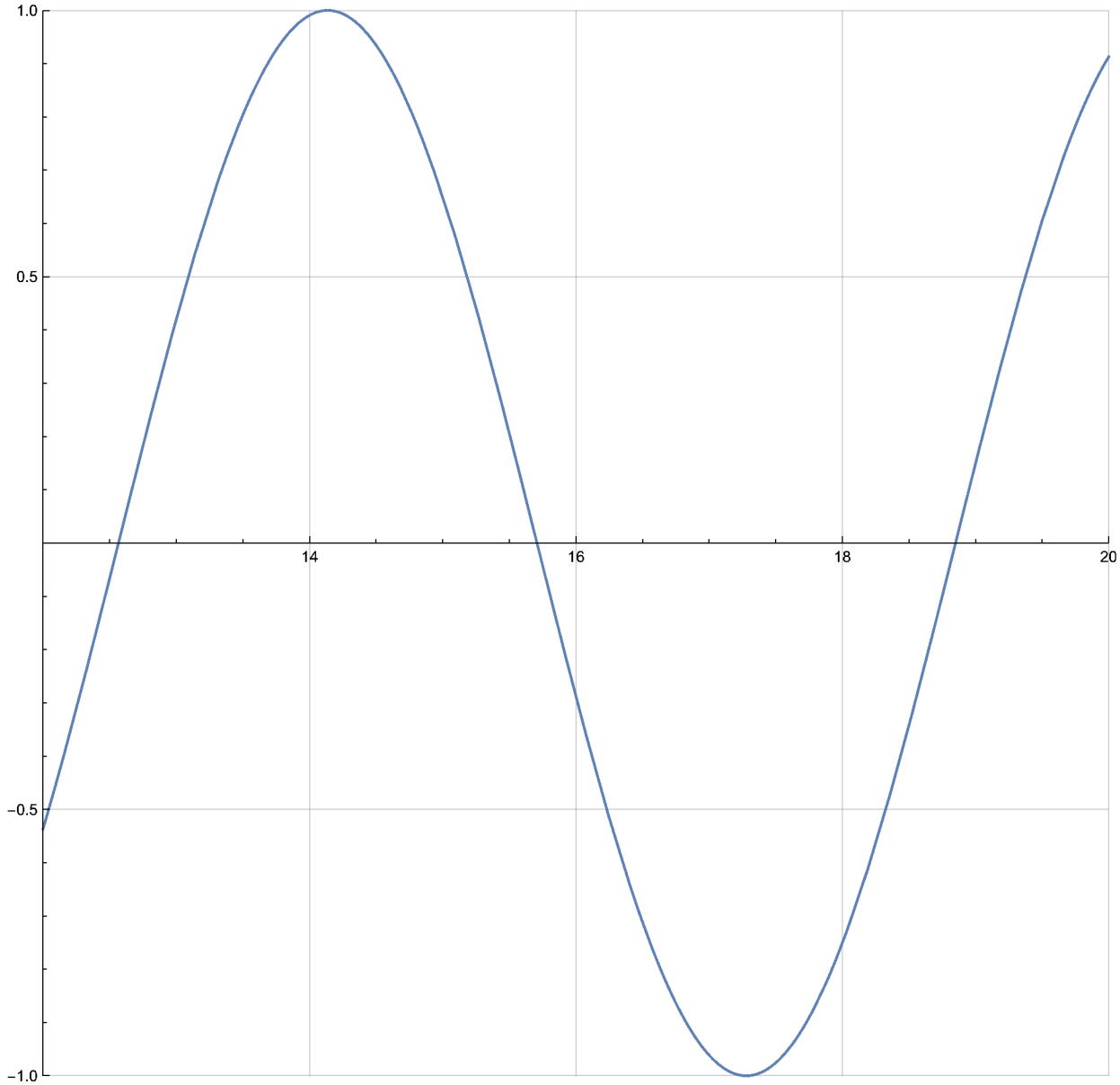
In[57]:=
```
f[x_] := Sin[x]
```

In[58]:=
```
Plot[f[x], {x, 12, 20}, PlotRange → {{12, 20}, {-1, 1}},
 GridLines → Automatic, AspectRatio → 1.0]
```

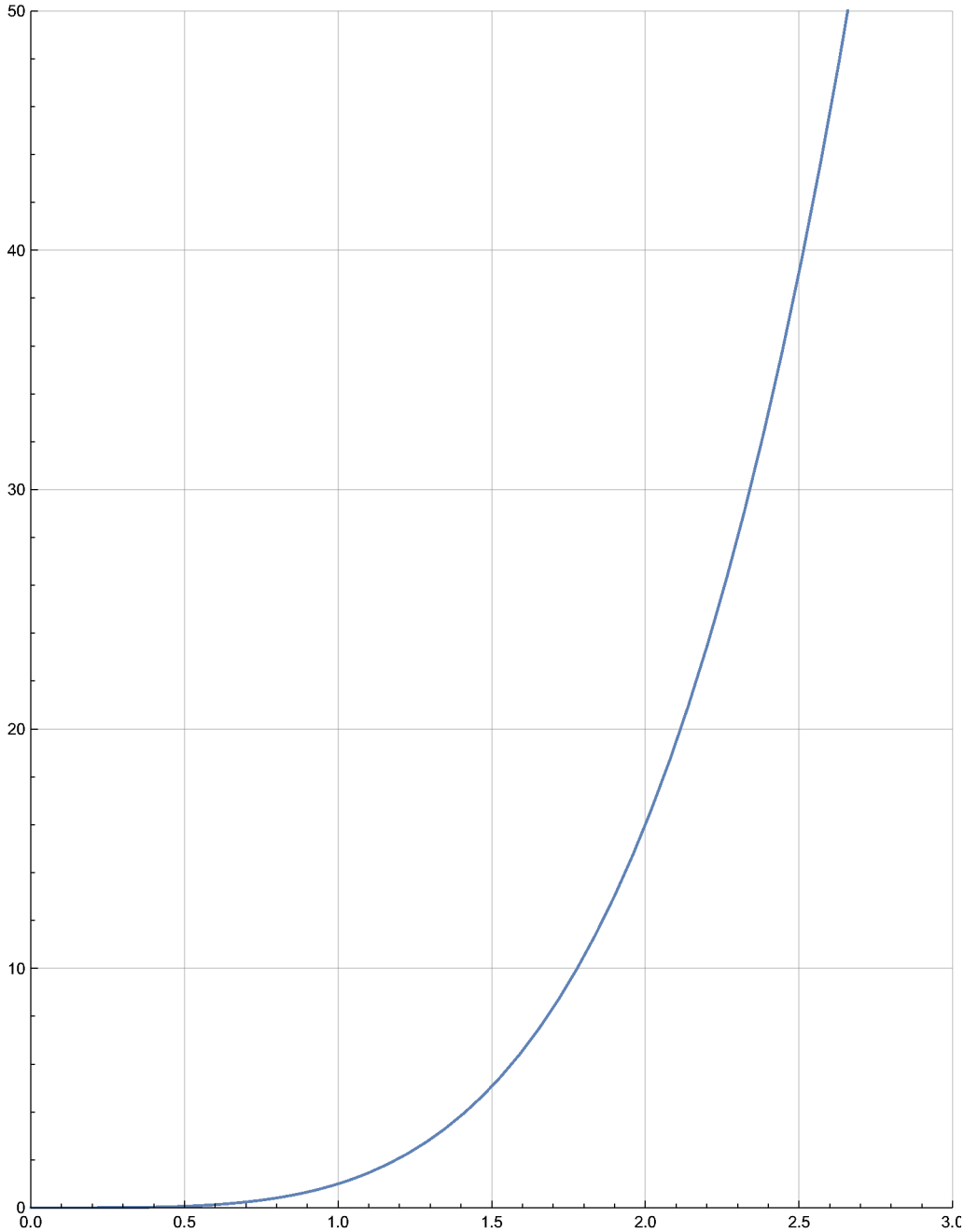
Out[58]=



In[59]:=
```
g[x_] := x^4
```

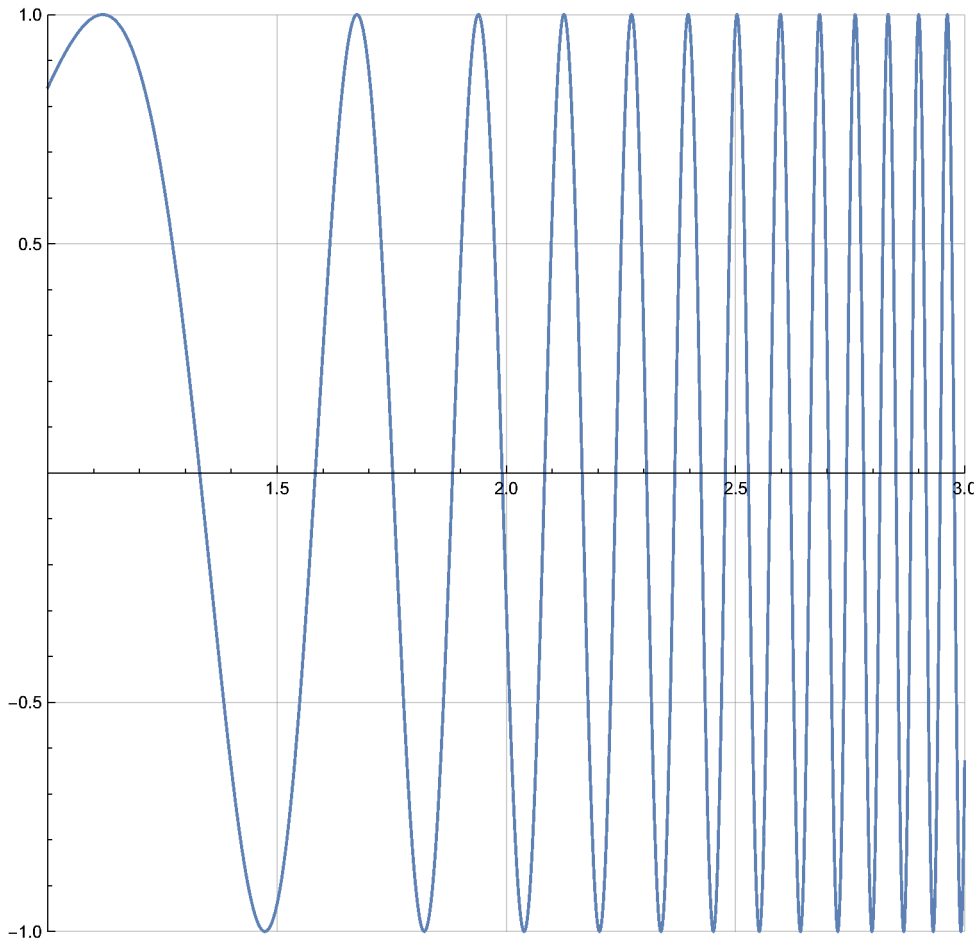
In[60]:=
```
Plot[g[x], {x, 0, 3}, PlotRange → {{0, 3}, {0, 50}},
 GridLines → Automatic, AspectRatio → 1.3]
```

Out[60]=



In[61]:=
```
Plot[f[g[x]], {x, 1, 3}, PlotRange → {{1, 3}, {-1, 1}},
 GridLines → Automatic, MaxRecursion → 15, AspectRatio → 1.0]
```

Out[61]=



Good Practice Problems for Problem 6

If you didn't find Problem 6 on the exam easy (that would be everybody), you should probably re-do all parts of Problem 3 of Spivak Chapter 5.

Here are a couple of other problems that use the same techniques as Problem 6....

For both of these, first, what is the limit? Second, for $\epsilon > 0$, what is the δ that keeps the function within ϵ of the limit.

In[62]:= $\lim_{x \to 1} \frac{1}{x^3}$

Out[62]= 1

In[63]:= With $a \neq 0$, $\lim_{x \to a} \frac{1}{x^2}$

Good Practice Problem for Problem 7

In problem 7, I tried for the easiest proof involving limits that I could come up with. The δ' that worked in the second limit was 2δ.

A slightly harder version of Problem 7 would be to show that the following two limits (if one exists), must be equal, and find the δ that makes it so.

So, assume this exists:

In[63]:= $l = \lim_{x \to 0} f(x)$

Out[63]= 0

Prove that this exists and is also l.

In[64]:= $\lim_{x \to 0} f(x^2)$

Out[64]= 0